Source ID	Search ID	Title	Author or Organisation	Year	Source Type	Link	Full Text Available?	Preliminary Decision	Reason
A-GS-01	A1	Save the Windows: A Residential Case Study Using Life Cycle Assessment	Katherine Switala-Elmhurst and Philip Udo-Inyang	2015	Journal article	https://www.proquest.com/openview/34e9550ecfa24c89a0ffb443283dbd00/1?pq-origsite=gscholar&cbl=5531556	No	Maybe	Highly relevant because it compares residential window restoration and replacement using lifecycle assessment, but the full text is not accessible. Retain temporarily and exclude during full-text screening if no accessible copy is found.
A-GS-02	A1	Methodology for Service Life Prediction of Window Frames	Daniel Fernandes, Jorge de Brito, and Ana Silva	2019	Journal article	https://cdnsciencepub.com/doi/abs/10.1139/cjce-2018-0453	No	Maybe	Directly relevant to window-frame service life, deterioration, and repair costs, but the full text is not accessible. Retain temporarily and exclude during full-text screening if no accessible version is found.
A-GS-03	A1	Sustainable Maintenance Strategy for Wood Windows Defects	Özlem Eren and Emine Merve Okumuş	2020	Journal article	https://dergipark.org.tr/en/pub/saufenbilder/article/687722	Yes	Keep	Examines window defects and how maintenance decisions can extend service life, which is directly relevant to maintenance, repair, and implementation conditions.
A-GS-04	A1	Analysis of the Inspection, Diagnosis, and Repair of External Door and Window Frames	A. Santos, M. Vicente, J. de Brito, I. Flores-Colen, and A. Castelo	2017	Journal article	https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29CF.1943-5509.0001095	No	Maybe	Examines defects, probable causes, inspection, and repair methods for external door and window frames. It is directly relevant to window failure and repair, but the full text is not currently accessible and the study also includes doors.
A-GS-05	A1	Serviceability fragility functions for New Zealand residential windows	David M. Carradine, Aman Kumar, Roger Fairclough, and Graeme Beattie	2020	Journal article	https://bulletin.nzsee.org.nz/index.php/bnzsee/article/view/1466	Yes	Maybe	Identifies damage and failure mechanisms in timber- and aluminium-framed residential windows, but the study focuses specifically on earthquake-related damage rather than ordinary maintenance, circularity, or service-life extension.
A-GS-06	A1	The Re-Seal Window: A Redesign of the Edge Seal of Insulated Glass Units to Facilitate Easy and Fast Remanufacturing	Juliette Mohamed	2020	Master's thesis	https://repository.tudelft.nl/file/File_24e03c8d-a38c-44e7-9414-d006bd36f7dc?preview=1	Yes	Keep	Directly examines how the weakest and shortest-life part of an insulated glazing unit could be replaced independently so that the longer-life glass and spacer components can remain in use.
A-GS-07	A1	Energy Simulation and Modeling for Window System: A Comparative Study of Life Cycle Assessment and Life Cycle Costing	Quddus Tushar, Muhammed A. Bhuiyan, and Guomin Zhang	2022	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0959652621041056	No	Exclude	Examines the thermal performance, embodied impacts, costs, glazing ratios, and frame materials of window systems, but does not address repair, component replacement, refurbishment, or service-life extension.
A-GS-08	A1	Steel Window Restoration: Effective Strategies for Restoring and Upgrading Historic Steel Windows	Arthur Femenella Sr. and Arthur Femenella Jr.	2025	Journal article / peer-reviewed practice article	https://www.jstor.org/stable/48853662	Yes	Keep	Provides practical evidence on the assessment, repair, restoration, upgrading, and possible replacement of steel windows, including the conditions under which restoration remains technically and economically realistic.
A-GS-09	A1	Economic Performance Assessment of Residential Building Retrofits: A Case Study of Istanbul	Ikbal Cetiner and Buket Metin	2017	Journal article	https://link.springer.com/article/10.1007/s12053-016-9501-4	No	Maybe	Examines lifecycle costs and economic decisions for residential retrofit measures, including window-system renewal. It may help explain why replacement options are selected, but it does not directly investigate window repair or component-level life extension.

A-GS-10	A1	Multi-Objective Optimization of Window Retrofit for Dormitory Buildings under Material Circularity Uncertainties: A Robust Surrogate-Assisted Framework	Yunze Shi, Xiaodong Cao, Zhaoyuan Yang, Meiyan Li, Xiyang Yu, and Xudong Yang	2026	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0360132326001782	No	Maybe	Examines lifecycle carbon, cost, comfort, and material-circularity trade-offs in window retrofit decisions. It is highly relevant to circularity uncertainty and lifecycle comparison, but it does not directly study repair or component replacement and uses a university dormitory case in China.
A-GS-11	A1	Life Cycle Perspectives of Fixed and Operable Wooden Windows	Dominika Búryová, Rozália Vaňová, Michal Gregor, Róbert Uhrín, and Pavol Sedlák	2025	Journal article	https://www.mdpi.com/2075-5309/15/24/4490	Yes	Keep	Compares the environmental impacts of wooden-window components and maintenance strategies, including glazing, hardware, coatings, and aluminium cladding. It is directly useful for identifying lifecycle trade-offs between longer service life, additional materials, and maintenance impacts.
A-GS-12	A1	Integrated Environmental, Energy and Cost Life-Cycle Analysis of Windows: Optimal Selection of Components	Shiva Saadatian, Nuno Simões, and Fausto Freire	2021	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0360132320308830	No	Maybe	Compares the environmental and economic impacts of glazing and frame alternatives across European climates. It may support the component and lifecycle analysis, but it does not directly examine repair, refurbishment, or component replacement.
A-GS-13	A1	Energy Performance and Economic Viability of Advanced Window Technologies for a New Finnish Townhouse Concept	Sudip Kumar Pal, Kari Alanne, Juha Jokisalo, and Kai Sirén	2016	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0306261915012830	No	Exclude	Focuses on the operational-energy performance and economic viability of advanced glazing technologies rather than repair, component replacement, refurbishment, or product-life extension.
A-GS-14	A1	Case Study on the Inspection and Repair of Window Condensation Problems in a New Apartment Complex	Sihyun Park and Seung-Yeong Song	2018	Journal article	https://ascelibrary.org/doi/abs/10.1061/(ASCE)CF.1943-5509.0001211	No	Maybe	Presents a real residential case in which window condensation defects were inspected and repair measures were developed. It is directly relevant to window failure, diagnosis, repair decisions, and the practical conditions that lead to intervention.
A-GS-15	A1	Life Cycle Evaluation of Smart Solar Windows Used in Buildings	Mohammad Hossein Jahangir and Rose Sadat Seyed Aboutorabi	2025	Book chapter	https://link.springer.com/chapter/10.1007/978-3-031-94422-2_5	No	Exclude	Focuses on the lifecycle costs, environmental impacts, and energy performance of smart-window technologies rather than repair, component replacement, refurbishment, or service-life extension.

A-GS-16	A1	Serviceability Fragility Functions for New Zealand Residential Windows	David M. Carradine, Aman Kumar, Roger Fairclough, and Graeme Beattie	2020	Journal article	https://d1wqtxts1xzle7.cloudfront.net/69140624/1387-libre.pdf?1631002604=&response-content-disposition=inline%3B+filename%3DServiceability_fragility_functions_for_N.pdf&Expires=1781778107&Signature=GArUOjt-3j4PLqKNjPCKkHVKT3bXTNCLqES7TT4YSLbV6dozLIRylspKyc1PTeJve-bvHKbs0Q5RjkPjbAzfOPJy5v~G9W-vZC9eA0zjAZqfh0fIXPYNovBu3A0BSr2exk3HsrIh7QOhQ5NuQJetBvR4-6DhWsVKMhr6mRPV2tmqZwZ34P6rS8FrwbEeEu69SHbOvshNVFhjZFaePlij6dUxbOhNgaGE4bWWjg3MHXo~bsaEt3EmzwPD0RMLMuI~sOtkFcJMZ9Jx5LDtKfIeMuw969y2VrOznmV24G6BdBMrjUPpitcBqOE6jjZF8u8qpdzlZzzeMNgn1QTqFAl2cw__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA	Yes	Exclude - Duplicate	Exact duplicate of A-GS-05.
A-GS-17	A1	Optimizing the Performance of Window Frames: A Comprehensive Review of Materials in China	Zhen Wang, Lihong Yao, Yongguang Shi, Dongxia Zhao, and Tianyu Chen	2024	Journal review article	https://www.mdpi.com/2076-3417/14/14/6091	Yes	Maybe	Reviews window-frame materials, structural designs, problems occurring during use, and sustainability considerations. It may support the analysis of material durability and design trade-offs, but it does not focus directly on repair, component replacement, or European window systems.
A-GS-18	A1	Life Cycle Assessment of a Wood-Aluminium Window According to European Norms	Michal Sečkár, Marián Schwarz, and Dušan Majer	2026	Journal article	https://www.nature.com/articles/s41598-026-46142-4	Yes	Keep	Provides a detailed lifecycle assessment of a European wood-aluminium window, including production, transport, use, end of life, and recycling. It identifies environmental hotspots that can support the lifecycle scenario comparison, although repair and individual component replacement are not included in the study.
A-GS-19	A1	Illustrative Case Study: Life Cycle Assessment of Four Window Alternatives	Mikołaj Owsianiak, Anders Bjørn, Heidi B. Bugge, Sónia M. Carvalho, Leise Jebahar, Jon Rasmussen, Caroline M. White, and Stig Irving Olsen	2018	Book chapter	https://link.springer.com/chapter/10.1007/978-3-319-56475-3_39	No	Maybe	Compares four window systems made from wood-composite, wood, wood-aluminium, and PVC and identifies environmental hotspots. It may support the lifecycle method and material comparison, but it does not directly examine repair, component replacement, or service-life extension.
A-GS-20	A1	Development of Designer Aids for Energy Efficient Residential Window Retrofit Solutions	Tim Ariosto, Ali M. Memari, and Ryan L. Solnosky	2019	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S221313881830359X	No	Exclude	Focuses mainly on add-on energy retrofit measures such as blinds, curtains, films, screens, shutters, and storm windows. It does not examine maintenance, repair, component replacement, or product-life extension in enough detail to answer the research question.

B-GS-01	B1	Interconnections: An Analysis of Disassemblable Building Connection Systems towards a Circular Economy	Timothy M. O'Grady, Roberto Minunno, Heap-Yih Chong, and Greg M. Morrison	2021	Journal article	https://www.mdpi.com/2075-5309/11/11/535	Yes	Keep	Examines how accessible and reversible connection systems can support the disassembly and reuse of building components. The case study is useful because it also identifies practical conflicts between accessibility, waterproofing, aesthetics, and construction requirements.
B-GS-02	B1	Design for Circular Disassembly of High-Performance Facade Systems	Naomi Galina Grigoryan and Alexandros Loutsioli Daskalakis	2025	Conference paper	https://www.proquest.com/openview/b154de70680e6a92c0a320676e5fc59b/1?pq-origsite=gscholar&cbl=1646340	Yes	Keep	Develops a design-for-disassembly method for high-performance façade systems and applies it to the Schüco AWS 75.SI+ window system. It is directly relevant to component access, disassembly planning, repair, refurbishment, and reuse.
B-GS-03	B1	Coding the Circularity: Design for the Disassembly and Reuse of Building Components	Salvatore Viscuso	2021	Journal article	https://re.public.polimi.it/handle/11311/1189127	Yes	Maybe	Examines how conditions for selective disassembly, remanufacturing, and component reuse can be included within computational and parametric design processes. The method is transferable, but the article does not focus specifically on windows or repair decisions.
B-GS-04	B1	Disassembly and Reuse of Demountable Modular Building Systems	Yang Yang, Calvin Luk, Bowen Zheng, Yifei Hu, and Albert Ping-Chuen Chan	2025	Journal article	https://ascelibrary.org/doi/abs/10.1061/JMENEA.MEENG-6243	Yes	Keep	Provides empirical evidence that designing a system for disassembly does not automatically make disassembly and reuse easy in practice. It identifies ownership, information tracking, inspection, repair, replacement, and value retention as important enabling conditions.
B-GS-05	B1	Robotic Disassembly and Window Upgrading	Begüm Aktaş and Mehrzad Esmaeili Charkhab	2024	Conference paper	https://tore.tuhh.de/dspace-cris-server/api/core/bitstreams/6d42b320-9975-43e6-9f77-d5baf510555a/content	Yes	Keep	Develops and tests a disassembly-map method for upgrading an aluminium-framed window by removing individual frame parts, gaskets, and glazing. It directly addresses selective disassembly, repair, replacement, upgrading, and window-life extension.
B-GS-06	B1	Design-for-Disassembly Principles for Apartment Blocks - Experimental Design in a Finnish Housing Project	Tuomo Joensuu and Arto Saari	2026	Journal article	https://www.sciencedirect.com/science/article/pii/S0959652626007195	Yes	Keep	Applies design-for-disassembly principles to a real Finnish apartment-building project and examines how general circular-design guidance can be translated into practical architectural and structural solutions. It is useful for identifying the design conditions, trade-offs, and implementation challenges that influence whether components can realistically be disassembled and reused.
B-GS-07	B1	Advancing Circular Economy Strategies in Modular Architecture through Design for Disassembly and Reuse	Kenza Belkhiri, Alina Mitache, and Viorel Ungureanu	2025	Journal article	https://potopk.com.pl/storage/im/2025/vol-2/no-2/articles/en/1029227im-2025-02-02-021__advancing-circular-economy-strategies-in-modular-architecture-through-design-for__ZWzGkJ.pdf	Yes	Maybe	Reviews how modularity, design for disassembly, standardisation, material passports, and component reuse can support circular construction. It also identifies regulatory, financial, technical, and supply-chain barriers, but it remains broad and does not focus specifically on windows or component-level repair.
B-GS-08	B1	A Selective Disassembly Multi-Objective Optimization Approach for Adaptive Reuse of Building Components	Benjamin Sanchez, Christopher Rausch, Carl Haas, and Rebecca Saari	2020	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0921344919305117	No	Maybe	Develops a method for selecting disassembly plans by considering physical, environmental, and economic constraints. The method is relevant to component access and reuse, but the full text is not openly available and the study concerns broader building-component recovery rather than window repair.

B-GS-09	B1	Increasing Reuse Potential by Taking a Whole Life-Cycle Perspective on the Dimensional Coordination of Building Products	Pieter Robert Beurskens and Elma Durmisevic	2017	Peer-reviewed conference paper	https://ris.utwente.nl/ws/portalfiles/portal/24832189/Conference_Proceedings_3rd_Green_Design_Conference_web.pdf	Yes	Keep	Examines how dimensional coordination and compatibility influence whether building products can be reused across more than one life cycle. It is directly relevant because reuse does not depend only on whether a component can be removed without damage, but also on whether its dimensions allow it to fit within a future building or product system.
B-GS-10	B1	Systems for Reuse, Repurposing and Upcycling of Existing Building Components	Colin Rose	2019	Doctoral thesis	https://discovery.ucl.ac.uk/id/eprint/10072754/	Yes	Keep	Examines the practical systems needed to identify, manage, reuse, repurpose, and upcycle existing building components. It is particularly relevant to information availability, component assessment, implementation barriers, and the conditions that determine whether reuse is realistic.
B-GS-11	B1	Taxonomy supporting design strategies for reuse of building parts in timber-based construction	Margherita Lisco and Radhlinah Aulin	2024	Journal article	https://www.emerald.com/ci/article/24/1/221/1230197/Taxonomy-supporting-design-strategies-for-reuse-of	Yes	Keep	Develops a clear taxonomy that connects design for disassembly and design for adaptability with the reuse of building components. It is useful because it clarifies overlapping circular-design concepts and explains how they can be applied during the design process.
B-GS-12	B1	Reversible Architecture: Reuse of Timber Building Parts in Circular Design	Margherita Lisco	2022	Licentiate thesis	https://lup.lub.lu.se/search/files/227764536/Thesis_Margherita_Lisco.pdf	Yes	Maybe	Examines circular design strategies, reversible connections, design for disassembly, adaptability, and the practical barriers that influence the reuse of timber building parts. It provides wider context and interview-based findings, but part of its evidence overlaps with B-GS-11.
B-GS-13	B1	Towards Zero-Waste Buildings: Building Design for Reuse and Disassembly	Václav Grmela	2020	Master’s thesis	https://odr.chalmers.se/server/api/core/bitstreams/b135bda8-64bd-4672-a7e9-5e3496ec9b59/content	Yes	Maybe	Explores practical architectural and component-design solutions for disassembly and future reuse. It may support the identification of design principles and visual examples, but it is a broad exploratory design thesis rather than a focused empirical study of window components.
B-GS-14	B1	Towards a Sustainable Circular Economy: Understanding the Environmental Credits and Loads of Reusing Modular Building Components from a Multi-Use Cycle Perspective	Yang Yang, Bowen Zheng, Calvin Luk, Ka-fai Yuen, and Albert Chan	2024	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S2352550924000526	Yes	Keep	Uses lifecycle assessment to examine the environmental benefits and burdens of reusing, repairing, replacing, recycling, and transporting modular building components across several use cycles. It is directly relevant to determining whether circular strategies create a real net environmental benefit.
B-GS-15	B1	Integrated Platform-Based Tool to Improve Life Cycle Management and Circularity of Building Envelope Components	Luca Morganti, Marco Demutti, Ioakeim Fotoglou, Eva Alessandra Coscia, Paolo Perillo, and Alessandro Pracucci	2023	Journal article	https://www.mdpi.com/2075-5309/13/10/2630	Yes	Maybe	Examines how digital information platforms can support the lifecycle management, traceability, reuse, and recycling of building-envelope components. It is useful for understanding information and infrastructure requirements, but it does not directly examine window repair or component replacement.
B-GS-16	B1	Design for Disassembly as an Alternative Sustainable Construction Approach to Life-Cycle Design of Concrete Buildings	Wasim Salama	2019	Doctoral thesis	https://repo.uni-hannover.de/items/64f4de0c-38e3-49e0-a2b6-4d0a2dd38605	Yes	Maybe	Examines design-for-disassembly principles, assessment factors, and practical design improvements for precast concrete buildings. It provides useful background on adaptability, connections, and component reuse, but it focuses on concrete construction rather than window or façade systems.

B-GS-17	B1	An ICT-Enabled Product Service System for Reuse of Building Components	David Ness, K. Xing, K. Kim, and A. Jenkins	2019	Conference paper	https://www.sciencedirect.com/science/article/pii/S2405896319311577	Yes	Keep	Examines how RFID, BIM, digital platforms, and product-service systems can support the identification, tracking, management, and reuse of building components. It is useful because it connects technical reuse potential with information, supplier responsibility, performance assurance, and business-model conditions.
B-GS-18	B1	Life Cycle Assessment of a Swedish Multifamily Building Designed for Disassembly and Flexibility: Impact of Allocation Methods on Future Scenarios	Sandra Moberg and Frida Görman	2025	Journal article	https://www.mdpi.com/2075-5309/15/17/3058	Yes	Keep	Compares linear and circular building scenarios and shows how lifecycle results change according to future-use assumptions and allocation methods. It is directly relevant to uncertainty, lifecycle trade-offs, and the risk of presenting future circular benefits as certain.
B-GS-19	B1	Metrics for Building Component Disassembly Potential: A Practical Framework	Havu Järvelä, Antti Lehto, Taika Pirilä, and Matti Kuittinen	2025	Journal article	https://research.aalto.fi/en/publications/metrics-for-building-component-disassembly-potential-a-practical-/	Yes	Keep	Develops and tests a streamlined method for assessing the disassembly potential of building components during early design. It is directly relevant to measurable design criteria, connection accessibility, material layers, and the development of a practical screening framework.
B-GS-20	B1	Disassembly Assessment Framework for Enhanced Reclamation of Façade Systems	Sarah Droste	2023	Master’s thesis	https://repository.tudelft.nl/file/File_6d358b1e-1d2f-4914-9b6c-22b018455c7a	Yes	Keep	Develops and demonstrates a façade-specific assessment framework combining disassembly potential, reclamation potential, disassembly time, material information, connection types, and end-of-life scenarios. It is directly relevant to the planned window-component map and preliminary screening framework.
C-GS-01	C1	Is Prolonging the Lifetime of Passive Durable Products a Low-Hanging Fruit of a Circular Economy? A Multiple Case Study	Mohamad Kaddoura, Marianna Lena Kambanou, Anne-Marie Tillman, and Tomohiko Sakao	2019	Journal article	https://www.mdpi.com/2071-1050/11/18/4819	Yes	Keep	Uses five real cases to compare the environmental and economic effects of extending the life of passive durable products through repair or refurbishment. It is directly transferable to windows because it shows that life extension is beneficial only when avoided production impacts remain greater than the impacts of spare parts, transport, repair, and supporting services.
C-GS-02	C1	Circular Economy Rebound	Trevor Zink and Roland Geyer	2017	Journal article	https://onlinelibrary.wiley.com/doi/abs/10.1111/jiec.12545	Yes	Maybe	Provides a foundational explanation of circular-economy rebound and shows how reuse, recycling, and other circular strategies may reduce the production of new goods by less than expected because they affect prices, demand, and markets.
C-GS-03	C1	The Circularity Paradox: The Rebound Effect, Governance, and the Limits of Corporate Sustainability	Taiz Viviane Dos Santos, Sandra Marilce Diavon Alvez, Simone Sehnem, and Dulcimar José Julkovski	2026	Journal review article	https://onlinelibrary.wiley.com/doi/full/10.1002/csr.70181	Yes	Maybe	Reviews how rebound effects appear within circular strategies and discusses the need for governance and demand-reduction measures. It is relevant conceptual evidence, but it overlaps with the stronger 2022 reviews and focuses more broadly on corporate sustainability than on product design.
C-GS-04	C1	The Rebound Effect of Circular Economy: Definitions, Mechanisms and a Research Agenda	Camila Gonçalves Castro, Adriana Hofmann Trevisan, Daniela C. A. Pigosso, and Janaina Mascarenhas	2022	Journal review article	https://www.sciencedirect.com/science/article/pii/S0959652622007685	Yes	Keep	Develops a conceptual framework that explains how circular rebound effects begin, develop, and may be mitigated. It is directly relevant to identifying plausible rebound pathways and distinguishing them from ordinary implementation problems.

C-GS-05	C1	How to Manage the Circular Economy Rebound Effect: A Proposal for Contingency-Based Guidelines	Pierluigi Zerbino	2022	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0959652622041567	No	Maybe	Directly relevant because it proposes practical guidelines for managing circular-economy rebound effects, but the full text is not currently accessible. Retain temporarily and exclude during full-text screening if an accessible copy cannot be obtained.
C-GS-06	C1	Evaluating Lifetime Extension Strategies to Close the Loop: Comparing the Material Circularity and Environmental Life Cycle Performance of Traction Batteries	Brianna Marie Hansen	2025	Master's thesis	https://unipub.uni-graz.at/obvugrhs/content/titleinfo/13352098	Yes	Maybe	Compares repair, reuse, remanufacturing, repurposing, and recycling through both lifecycle assessment and material-circularity indicators. The method is useful for understanding when lifetime extension creates a real environmental benefit, but the study focuses on traction batteries rather than building components and does not directly examine rebound effects.
C-GS-07	C1	Impacts, Synergies, and Rebound Effects Arising in Combinations of Product-Service Systems (PSS) and Circularity Strategies	David Sarancic, Julija Metic, Daniela C. A. Pigosso, and Tim C. McAloone	2023	Peer-reviewed conference article	https://www.sciencedirect.com/science/article/pii/S2212827123000963	Yes	Keep	Examines how different combinations of product-service systems and circular strategies may create environmental, social, and economic benefits, but also rebound effects. It is useful because it applies a rebound-effect framework during early design rather than assuming that combining several circular strategies will automatically improve sustainability.
C-GS-08	C1	Careless Product Use in Access-Based Services: A Rebound Effect and How to Address It	Laura Ackermann and Vivian S. C. Tunn	2024	Journal article	https://www.sciencedirect.com/science/article/pii/S0148296324001474	Yes	Keep	Demonstrates how a circular business model intended to increase product use and extend product life can produce the opposite result when users take less care of products they do not own. It is directly relevant to behavioural rebound, user acceptance, and the practical conditions needed for product-life extension.
C-GS-09	C1	A Study of the Rebound Effect on the Product-Service System: Why Should It Be a Top Priority?	Salman Alfarisi, Yuya Mitake, Yusuke Tsutsui, Hanfei Wang, and Yoshiki Shimomura	2022	Peer-reviewed conference article	https://www.sciencedirect.com/science/article/pii/S2212827122006953	Yes	Maybe	Proposes a lifecycle-based framework for identifying the types, causes, and timing of rebound effects within product-service systems. The method is relevant, but its conceptual contribution overlaps with stronger rebound frameworks already retained from Search C.
C-GS-10	C1	Towards Sustainable Development through the Circular Economy-A Review and Critical Assessment on Current Circularity Metrics	Blanca Corona, Li Shen, Denise Reike, Jesús Rosales Carreón, and Ernst Worrell	2019	Journal review article	https://www.sciencedirect.com/science/article/pii/S0921344919304045	Yes	Keep	Critically reviews circularity metrics and shows that measuring material circulation alone may hide environmental, economic, or social burden shifting. It is directly relevant to distinguishing genuine environmental improvement from circularity claims that do not account for wider lifecycle consequences.
D-GS-01	D1	The Influence of Durability and Recycling on Life Cycle Impacts of Window Frame Assemblies	Stephanie Carlisle and Elizabeth Friedlander	2016	Journal article	https://link.springer.com/article/10.1007/s11367-016-1093-x	Yes	Keep	Compares the lifecycle impacts of aluminium, wood, aluminium-clad wood, and PVCu window frames under different maintenance, component-replacement, service-life, and end-of-life scenarios. It is directly relevant to the 50-year scenario comparison because it shows how assumptions about durability and maintenance can substantially change the environmental result.
D-GS-02	D1	Maintaining or Replacing a Building's Windows: A Comparative Life Cycle Study	Liza Sällström Eriksson and Sofia Lidelöw	2025	Journal article	https://www.emerald.com/ijbpa/article/43/4/766/1257587	Yes	Keep	Directly compares window maintenance and replacement using lifecycle climate assessment and lifecycle costing over a 60-year period. It is highly relevant because it examines the trade-off between avoided material production and the operational-energy savings associated with replacement.

D-GS-03	D1	A Comparative Life Cycle Study of Window Interventions: Impact of Building Characteristics and Local Context	Liza Sällström Eriksson and Sofia Lidelöw	2025	Journal article	https://www.sciencedirect.com/science/article/pii/S2352710225012562	Yes	Keep	Compares maintenance and replacement scenarios for windows in two residential buildings and examines how building characteristics, location, heating emissions, transport, and other assumptions change the preferred intervention. It is directly relevant to lifecycle uncertainty and contextual conditions.
D-GS-04	D1	Life Cycle Assessment of Buildings and City Quarters Comparing Demolition and Reconstruction with Refurbishment	Verena Weiler, Hannes Harter, and Ursula Eicker	2017	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0378778816314141	No	Exclude	Compares refurbishment with demolition and reconstruction at whole-building and urban-quarter scale. Although the lifecycle method is relevant in general, it does not provide window-specific evidence on repair, component replacement, or window service-life extension.
D-GS-05	D1	Comparative Life Cycle Assessment of Reconstruction and Renovation for Carbon Reduction in Buildings	Hyojin Lim	2025	Journal article	https://www.mdpi.com/2075-5309/15/18/3388	No	Exclude	Compares whole-building renovation, reconstruction, and continued use. Windows are included only as one part of a larger renovation package, so the study cannot provide specific evidence about window maintenance, repair, component replacement, or full-window replacement.
D-GS-06	D1	Cradle-to-Grave Life-Cycle Assessment Within the Built Environment: Comparison Between the Refurbishment and the Complete Reconstruction of an Office Building in Belgium	Anne-Françoise Marique and Barbara Rossi	2018	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0301479718301622	No	Exclude	Compares refurbishment with complete demolition and reconstruction at whole-building scale. Although the lifecycle approach is relevant, it does not provide specific evidence on window repair, refurbishment, component replacement, or full-window replacement.
D-GS-07	D1	Save the Windows: A Residential Case Study Using Life Cycle Assessment	Katherine Switala-Elmhurst and Philip Udo-Inyang	2015	Journal article	https://www.proquest.com/openview/34e9550ecfa24c89a0ffb443283dbd00/1?pq-origsite=gscholar&cbl=5531556	No	Exclude - Duplicate	Exact duplicate of A-GS-01. It directly compares residential window restoration and replacement through lifecycle assessment, but the full text is not accessible.
D-GS-08	D1	Comparative Whole-Building Life Cycle Assessment of Renovation and New Construction	Vaclav Hasik, Elizabeth Escott, Roderick Bates, Stephanie Carlisle, Billie Faircloth, and Melissa M. Bilec	2019	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0360132319304287	Yes	Maybe	Proposes a clear method for comparing renovation with new construction while accounting for retained components and their later maintenance, repair, replacement, and end-of-life impacts. The method is transferable to the window scenario comparison, but the study itself operates at whole-building scale.
D-GS-09	D1	Embodied Life Cycle Assessment (LCA) Comparison of Residential Building Retrofit Measures in Atlanta	Arezoo Shirazi and Baabak Ashuri	2020	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S0360132320300020	Yes	Maybe	Compares embodied impacts and operational-energy payback for residential retrofit measures, including window upgrading. It may provide supporting evidence on the initial environmental burden of window replacement, but it does not compare window maintenance or component repair with full replacement.
D-GS-10	D1	Life Cycle Operating Energy Saving from Windows Retrofitting in Heritage Buildings Accounting for Technical Performance Decay	Giovanni Litti, Amaryllis Audenaert, and Monica Lavagna	2018	Journal article	https://www.sciencedirect.com/science/article/abs/pii/S235271021830007X	Yes	Keep	Directly compares window interventions ranging from basic maintenance to complete replacement and examines how component deterioration, service life, and repeated interventions change long-term performance. It is useful for defining realistic scenario assumptions, although it assesses operational energy rather than a complete environmental lifecycle.